

Full Marks: 60

Time: 3Hrs

Question 1 to 10 carries 1 mark each.

- ① State True/False: Rotating an object about the origin in 2D is similar to rotation about z-axis in 3D
- ② The left (L bit) bit of the region code of a point (X,Y) is '1' if
a) $X > X_{WMIN}$ b) $X < X_{WMIN}$ c) $X < X_{WMAX}$ d) $X > X_{WMAX}$
- ③ A 3D representation of an object through linear, circular or some other representation is called
a) Quadric surface b) Sweep representation c) Torus d) None of these
4. When projection is perpendicular to the view plane, we have _____ projection.
5. During processing of pairs of polygon vertices for polygon clipping, if the first vertex is inside and the second vertex is outside the window boundary then
A) Save both the intersection point with the window boundary and the second vertex
B) Save the edge intersection with the window boundary
C) Both A and B D) None of the above
6. During construction of an axonometric projection, the following transformation(s) is/are used:
a) Rotation b) Translation c) Scaling d) a & b e) b & c
- ⑦ Give the formula for the foreshortening factor in axonometric projection.
- ⑧ When the projection angle, $\alpha = 90^\circ$, the resulting projection is:
a) Perspective b) Cavalier c) Cabinet d) orthographic
- ⑨ Distance from centre of projection to projection plane is _____ for perspective projection.
- ⑩ Origin is affected by which transformation?
- ⑪ In perspective projection, suppose the projection plane is parallel to y axis. What happens to the set of parallel lines that are parallel to say axis? [2]
- ⑫ What are the two approaches for removing hidden surface problems? Explain. [4]
- ✓ 13. Use the Cohen-Sutherland line clipping algorithm to clip line P1 (70, 20) and P2(100,10) against the window with coordinates (50,10), (80, 40). Give all the steps of your solution and finally quote the coordinate end points of the clipped line. [8]
- ✓ 14. Find the cubic Bezier curve for different values of u at intervals of 0.2 for the following 4 control points: A (2,4), B (4,7), C (7,2) and D (9,6). [10]
- ⑮ Find the orthographic projection of a unit cube onto the $x=0$ and $z=0$ planes. [5]
- ⑯ Write the function to generate a symmetric fractal. [5]
- ✓ 17. Derive the transformation matrix for rotating any object by 45° about an axis passing through the origin and point (10,10,10). $\left(\frac{10}{\sqrt{200}} = 0.707\right)$ [10]
- ✓ 18. Explain perspective projection and the different types of perspective projection with proper diagrams.

OR

Give the steps for designing an Animation sequence. [6]

